

Mobile Development Final Report

1. Introduction

1.1 Project Overview

I designed the application as a comprehensive platform for browsing and purchasing perfumes. It features a dynamic product listing, detailed product pages, a shopping cart, and an order management system. I leveraged custom state management and integrated several third-party libraries to provide a seamless user experience.

2. Features

2.1 Navigation

- **React Navigation:** I implemented a robust navigation system using React Navigation, including stack and drawer navigations.
- **Dynamic Drawer Content:** I created custom drawer navigation that dynamically lists product categories, allowing users to filter perfumes based on selected categories.
- **Deep Linking:** I enabled users to directly navigate to specific sections within the app through deep linking.

2.2 Theming

- **Dark and Light Theme Support:** The app adapts to the system's theme preferences using the `useColorScheme` hook, ensuring a consistent user experience across different themes.
- **Custom Theme Provider:** I integrated `react-native-paper` for theming, providing a cohesive and modern UI.

2.3 State Management

- **Custom State Management:** I implemented a custom state management solution using the React Context API and a reducer to manage global states like the cart, user data, and orders.

- **Asynchronous State Updates:** The state is persisted using AsyncStorage, ensuring that cart and order data is maintained across app sessions.

2.4 Local Storage

- **AsyncStorage for Persistence:** I utilized AsyncStorage to store and retrieve cart items and orders, allowing users to resume their shopping experience even after closing the app.
- **Order Management:** Completed orders are saved locally, enabling users to view their order history.

2.5 Shopping Cart Functionality

- **Add to Cart:** I implemented functionality that allows users to add perfumes to their cart, with quantities and total prices dynamically updated.
- **Remove from Cart:** I ensured that items can be removed from the cart, with the UI and state updated in real-time.
- **Proceed to Checkout:** I developed a complete checkout process that simulates order placement, clears the cart, and saves the order locally.

2.6 Product Display and Interaction

- **Product Listing:** I displayed perfumes in a grid format, featuring images, names, and prices.
- **Product Details:** I provided a detailed view of each perfume, including a gallery, description, and price, with the option to add to the cart.
- **Product Carousels:** I implemented a carousel to feature highlighted perfumes and testimonials, enhancing the visual appeal.

2.7 Testimonials and User Interaction

- **Testimonials Carousel:** I added a carousel to display user testimonials, providing social proof and enhancing user engagement.
- **Interactive UI Elements:** I ensured that users can interact with various UI elements, such as pressing buttons to add items to the cart or navigating between screens.

2.8 Order Management

- **Order History:** I enabled users to view a history of their completed orders, which is persisted using AsyncStorage.
- **Order Details:** I ensured that each order includes a list of purchased items, the total amount, and the date of purchase.

3. Concept Development

3.1 Initial Idea

I envisioned the project as an e-commerce platform specifically focused on perfumes. My goal was to design an intuitive and visually appealing interface that would allow users to easily browse through products, view detailed information, and make purchases.

3.2 Research and Analysis

I conducted research on existing e-commerce platforms, with a particular focus on cosmetics and fragrance apps. I identified key features as essential, including product galleries, detailed product views, and efficient cart management.

3.3 Feature List

- **Product Listing:** I planned to display all available perfumes in a grid format.
- **Product Details:** I aimed to provide detailed information about each perfume, including a gallery, pricing, and descriptions.
- **Shopping Cart:** I designed functionality to allow users to add items to their cart, view totals, and remove items.
- **Checkout Process:** I simulated the process of placing an order.
- **Drawer Navigation:** I incorporated easy access to categories and other screens through a drawer menu.
- **Order Management:** I enabled users to view and manage their orders.
- **Theme:** I made it easy to change the color across the app, which reflects instantly.

4. Wireframing

4.1 Wireframe Designs

I created wireframes to establish the layout of the application. Key screens such as the product list, product details, cart, and checkout were sketched to ensure a user-friendly flow. These wireframes can be found in the `/wireframes` folder.

4.2 Feedback and Iteration

During the wireframing process, I received feedback and made iterations. I improved the product detail page to better display images and descriptions and streamlined the checkout process.

5. Prototyping

5.1 Prototype Development

I developed the prototype using React Native and Expo. My initial focus was on building functional screens with basic navigation and state management. I progressively added styling and UI enhancements.

5.2 User Testing

I conducted user testing to evaluate the app's usability. Testers were asked to navigate through the app, add items to the cart, and complete a purchase. The feedback was positive, with some suggestions for minor improvements in navigation and UI.

6. Development

6.1 Tools and Technologies

- **React Native & Expo:** I used these as the core framework and platform for building and running the application.
- **React Navigation:** I managed navigation between screens using this, including drawer and stack navigations.
- **AsyncStorage:** I utilized this for persisting cart data and managing orders locally.
- **Axios:** I used this for API interactions and managing backend communication.

6.3 Challenges Faced

- **State Management:** Managing the cart state across different screens and ensuring accurate updates in the UI was challenging, but I successfully implemented it.
- **AsyncStorage Management:** Ensuring data persistence across app restarts required me to carefully handle AsyncStorage.
- **Drawer Navigation:** Integrating custom drawer content to handle category-based navigation was complex, but I ultimately succeeded.

6.4 Key Features Implementation

- **Product Listing:** I implemented this using `FlatList` to efficiently display products in a grid.
 - **Cart Management:** I used a combination of state management and AsyncStorage to handle cart items.
 - **Drawer Navigation:** I customized drawer navigation to dynamically list product categories and handle navigation based on selected categories.
 - **Checkout Process:** I simulated order creation and stored completed orders using AsyncStorage.
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Appendix

A.1 Code Snippets

Include significant code snippets that demonstrate key implementations, such as the cart management logic, navigation setup, and AsyncStorage handling.

A.2 Additional Resources

- **React Native Documentation:** <https://reactnative.dev/docs/getting-started>
- **Expo Documentation:** <https://docs.expo.dev/>
- **React Navigation:** <https://reactnavigation.org/docs/getting-started>

A.3 Inspiration

- <https://www.amazon.com/>

- <https://3saf.com/en/>